

Nacre-Inspired Metal-Organic Framework Coatings Reinforced by Multiscale Hierarchical Cross-linking for Integrated Antifouling and Anti-Microbial Corrosion

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A long-standing quest in marine materials science has been the development of tough and effective antifouling coatings for diverse surface protection. However, most commercial coatings are severely limited by poor mechanical behavior and unsustainable passive biocidal effect, leading to irreversible marine biofouling and even microbiologically influenced corrosion (MIC). Herein, inspired by the amorphous/crystalline feature within nacreous platelets, a mechanically robust antifouling coating composed of biopolymer-based hydrogel and dense metal-organic frameworks (MOFs) is developed. Tailoring the cross-linked networks across multiscale interfaces can furnish strength, dissipate strain, and improve toughness of the building blocks, resulting in a firm and scalable configuration on various substrates regardless of material category and surface topology. The resultant coating as a suitable reservoir exhibits a unique active defensive behavior of intelligent MOF degradation or drug release, enabling a groundbreaking performance for broad-spectrum biofouling and corrosion control. Notably, neither attachment of marine organisms nor MIC of metal substrates is observed and aggravated during the prolonged testing process in complex biological environments. This study provides distinctive insights into the underlying multimechanisms of comprehensive anti-fouling-corrosion and pioneer a rational strategy to design next-generation reliable MOFs-derived coatings in marine environments.

global challenge for our maritime affairs.^[1,2] Fouling organisms that settle on underwater surfaces enhance hydrodynamic drag, hamper ship maneuverability, increase fuel consumption, and weaken instrument function, bringing seriously adverse effects on economic maintenance and environmental safety.^[3] Antifouling coatings are regarded as one of the most effective and feasible approaches to address biofouling issues. Driven by the restrictive evaluation and regulatory regime on the use of biocides, notably the legislation that tributyltin was forbidden due to its high teratogenicity, significant efforts have been devoted to searching for eco-friendly and efficient marine antifouling coatings.^[4–7] Although some synthetic coatings, such as self-polishing coating,^[6] superwetting coating,^[9] dynamic biodegradable coating,^[10] and so forth, can endow surfaces with antifouling capacity through the antifoulant, surface energy, topography, modulus, or strong hydration layers, it is unlikely that the component design based on one attribute will be sufficient to combat biofouling in the harsh marine environment. In particular, they are only capable of partial retention of biofilms composed of

encapsulated microbes and matrices of secreted extracellular polymeric substances, instead of thoroughly cleaning them out.^[11,12] Eventually, with accumulated organic pollutants as

1. Introduction

Marine biofouling, the undesired colonization by which marine organisms attach to submerged structures, represents a foremost

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The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/adfm.202305995>

DOI: 10.1002/adfm.202305995

nutrients, the coating surfaces become easy breeding grounds for algae and invertebrates, which causes secondary marine biofouling.^[13] Furthermore, mature biofilm formation shows environmental gradients in oxygen and pH and capture metal compounds from the materials as resources through extracellular electron transfer, resulting in the occurrence of MIC.^[14,15] Unfortunately, current antifouling coatings cannot inhibit the corrosion process that requires additional and complex physical barriers against the penetration of corrosive species. Hence, it is highly sought after to develop integrated antifouling and anti-corrosion coatings that can remove fouling organisms effectively as well as interrupt MIC by killing the attached biofilm completely.

MOFs are an emerging class of crystalline coordination materials consisting of metal clusters or metal ions and organic linkers.^[16,17] Intrinsic porous characteristics, well-dispersed active centers, and abundant functionalities enable MOFs to be promising candidates in chemistry, energy, environment, and biomedicine.^[18–20] Recently, successful applications have been extensively investigated in preventing bacterial attachment, inhibiting bacterial production, and killing bacteria.^[21] The extraordinary designability of MOFs offers us an opportunity to optimize the antimicrobial performance at the molecular level by rationally selecting metal ions/clusters or organic linkers, which is considered as a significant competitive advantage of MOFs over traditional bacterial-killing materials.^[22] Besides, owing to their diverse spatial structure and tunable composition, MOFs play a crucial role in universal carriers of targeted drugs to obtain multiple bioactive abilities.^[23] The released metal ions, linkers, and drugs can interact with the bacterial cell membrane, disturb metabolic activities, cause leakage of cell components, and finally kill microorganisms.^[23,24] Especially the adjustable release process can be performed by regulating the structure stability of MOFs from their crystalline, size, and building units, leading to lower toxicity, longer valid time, and success in avoiding endogenous accumulation.^[25] Thus, MOFs exhibit a high potential for alleviating marine biofouling and MIC deterioration. Substrate-supported heteroepitaxial synthesis is the most straightforward route to construct MOFs-based coatings composed of an uninterrupted polycrystalline layer.^[26] Nonetheless, on the one hand, the formed thin film is not mechanically robust enough for scale-up production and practical utility. On the other hand, ineffective control over crystal growth and defects conducts unsatisfactory antifouling performance.^[26,27] Nowadays, the most universal strategy for improving MOFs applicability is to use the principle of physical blending that MOFs are embedded in a polymeric matrix with good processability.^[28] Unfortunately, mixed-matrix coatings are limited by easy aggregation, poor dispersion, and weak interfacial interactions. More importantly, contacting components with fouling organisms are still dominated by the polymer matrix with low antifouling activity. Therefore, a better molecular-level integration of mutually exclusive rigid MOFs and flexible matrix is urgently required to feed the demand for multifunctional coatings in combination with strong mechanical property and intriguing antimicrobial activity.

Biological organisms have tailored self-adapting design principles to obtain their respective functionalities through evolution of millions years. Marine mollusks provide an ideal case for assembling complex organic/inorganic components under mild con-

ditions to create hierarchical and heterogeneous nanocomposites with the ordered distribution.^[29,30] Depending on the typical biomineralization process, the natural nacre as a product of mollusks is equipped with a highly regular “brick-and-mortar” organization of inorganic aragonite platelets ($\approx 95\%$) connected by a thin organic interlayer ($\approx 5\%$).^[31] Such a configuration not only ensures adequate interlayer adhesion and interaction but also exhibits exceptional mechanical performance to protect the internal soft body.^[32] Given this intriguing phenomenon, we believe that the biomimetic mineralization-induced process might suggest an efficient strategy to construct advanced MOFs-based coatings with comprehensive anti-fouling-corrosion properties and mechanical robustness. Despite its promise, such an approach has yet to be considered or achieved in the creating marine antifouling materials.

Herein, we report a hydrogel-anchored MOF-based coating (HMC) that imitates nacreous amorphous/crystalline heterophase structures to enhance the physicochemical stability and strengthen its anti-fouling-corrosion performance (**Figure 1a**). The HMC is achieved by assembling multiscale building blocks in an in situ penetrated biomineralization fashion. These building blocks are composed of nanoscale reinforcing and antifouling MOF units glued by sophisticated hydrogels containing sodium alginate (SA), carboxymethyl cellulose (CMC), and polyethyleneimine (PEI). Concretely, SA provides abundant MOF nuclei, CMC improves the material flexibility, while PEI enhances the interfacial binding force with substrates and eliminates the surface charge effect. Encouraged by the proper combination of hierarchical topography and chemical compositions highly resembling the natural nacre, the resultant HMC shows not only a surface-independent adhesion but also a mechanically robust anti-fouling-corrosion ability. The synthesized HMC could be facilely and uniformly coated on diverse target substrates such as metal, rubber, glass, and plastic without arduous surface pretreatment. The as-prepared HMC demonstrates weak attraction and superior resistance to proteins, bacteria, and algae in complex biological environments. Benefitting from the boosting antibacterial and antibiofilm activities, the MIC of metallic materials is also effectively decayed by the coating (**Figure 1b**). Significantly, the large surface area and high porosity of HMC pose a versatile platform for drug loading to further promote the protection level in demand. This work has provided a paradigm for MOF-based coatings to build an efficient bridge between artificially modified function and mechanical performance, which unlocks engineering possibilities for marine antifouling and anti-corrosion applications.

2. Results and Discussion

2.1. Fabrication and Characterization of the HMC

The HMC was fabricated through a hydrogel-anchored painting method under mild conditions, referring to a two-step technology of stably sticking anti-fouling-corrosion MOF-based layers on solid surfaces (**Figure 2a**). Typically, as a soft biopolymer-based phase, an extremely tough hydrogel was first constructed with the well-defined thickness on the substrate. SA, CMC, and PEI with hydrogen bonding and covalently cross-linked networks were used as precursor slurry to mimic multiscale moieties in

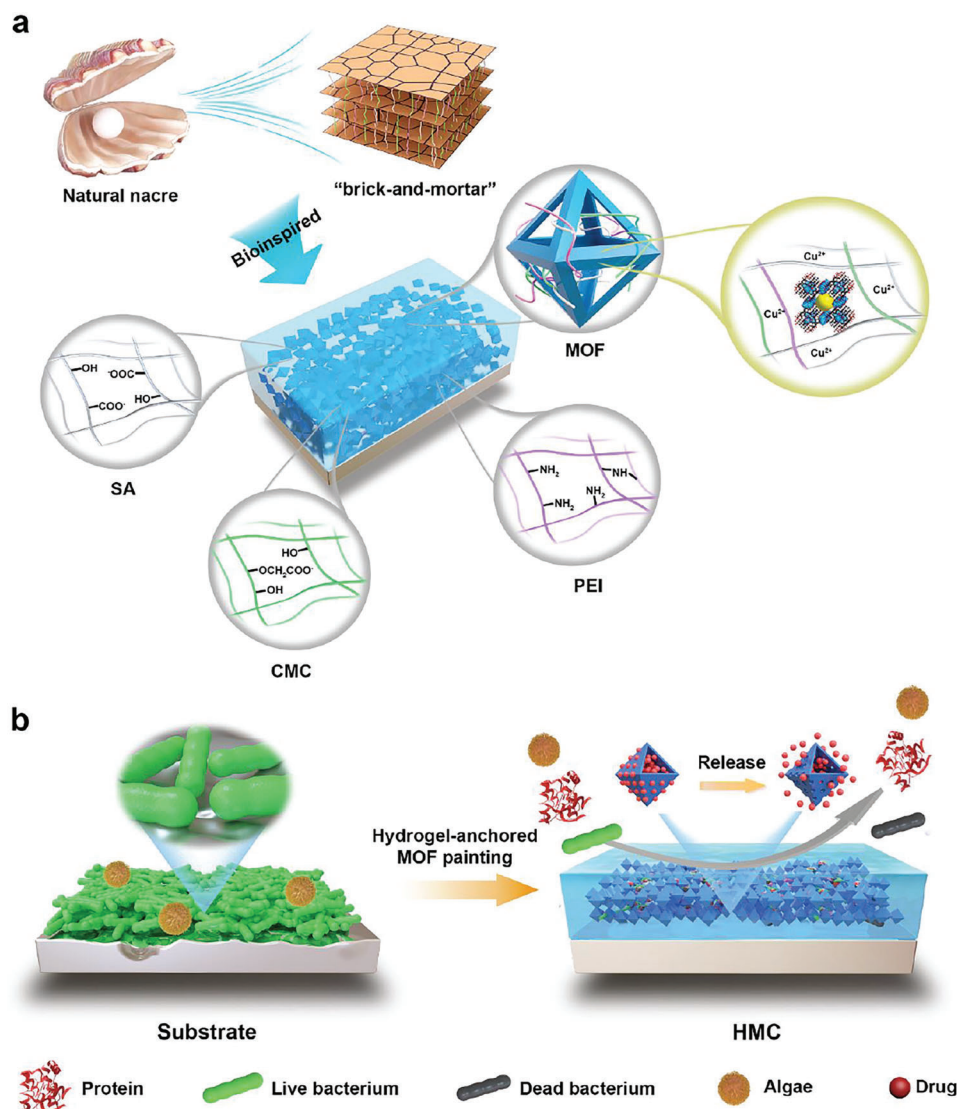


Figure 1. Schematic illustration of the HMC with anti-fouling-corrosion properties. a) The well-organized microstructure of the nacre-inspired HMC featuring crystalline MOFs and amorphous hydrogel with multiscale crosslinked interactions. b) Antifouling and anti-corrosion mechanism of the HMC on target substrates via a hydrogel-anchored MOF painting technique.

natural nacre such as soft β -chitin fibril and the analogous silk protein.^[33] Divalent metal ions such as copper ions (Cu^{2+}) that can bind preferentially to the guluronic acid blocks of alginate chains in a highly synergetic manner are incorporated into the hydrogel to further improve the mechanical behaviors via ionic crosslinks. Undoubtedly, the resultant hydrogel featured high-density crosslinking interfaces at multiple scales, acting as a confined nucleation reactor for subsequent reactions. Then, targeting the metal ions in alginate, the in situ biomineralization process was readily mediated by depositing organic linkers in the hydrogel system, leading to nanoscale surface-attached MOF heterophase units. The network entanglements from the coordination and confinement effects guarantee a stable bonding between the hydrogel and the MOFs. Finally, the nacre-inspired HMC with a highly ordered structure was intactly generated on the surface.

Taking 304 stainless steel (304 SS) as an exemplified substrate material, the scanning electron microscopy (SEM) images demonstrate the stepwise preparation of the HMC composed of the hydrogel as the organic framework and the MOFs as the inorganic component (Figure 2b–d). The original 304 SS is polished to warrant a smooth and clean surface for spreading or spraying operations (Figure 2b). Similar to the mortar structure of the nacre, the ternary hydrogel coating is further paved on the surface to form uniform and dense three-dimensional networks (Figure 2c). According to Pearson's hard/soft acid/base principle, brick-like HKUST-1 could be in situ synthesized in the hydrogel system upon immersing the coated substrates in the reaction solution of carboxylate-based ligands.^[34] In comparison, few polycrystalline HKUST-1 nanoparticles adhere directly to the uncoated 304 SS, suggesting that the presence of a hydrogel layer provides the basis for MOF growth (Figure S1, Supporting

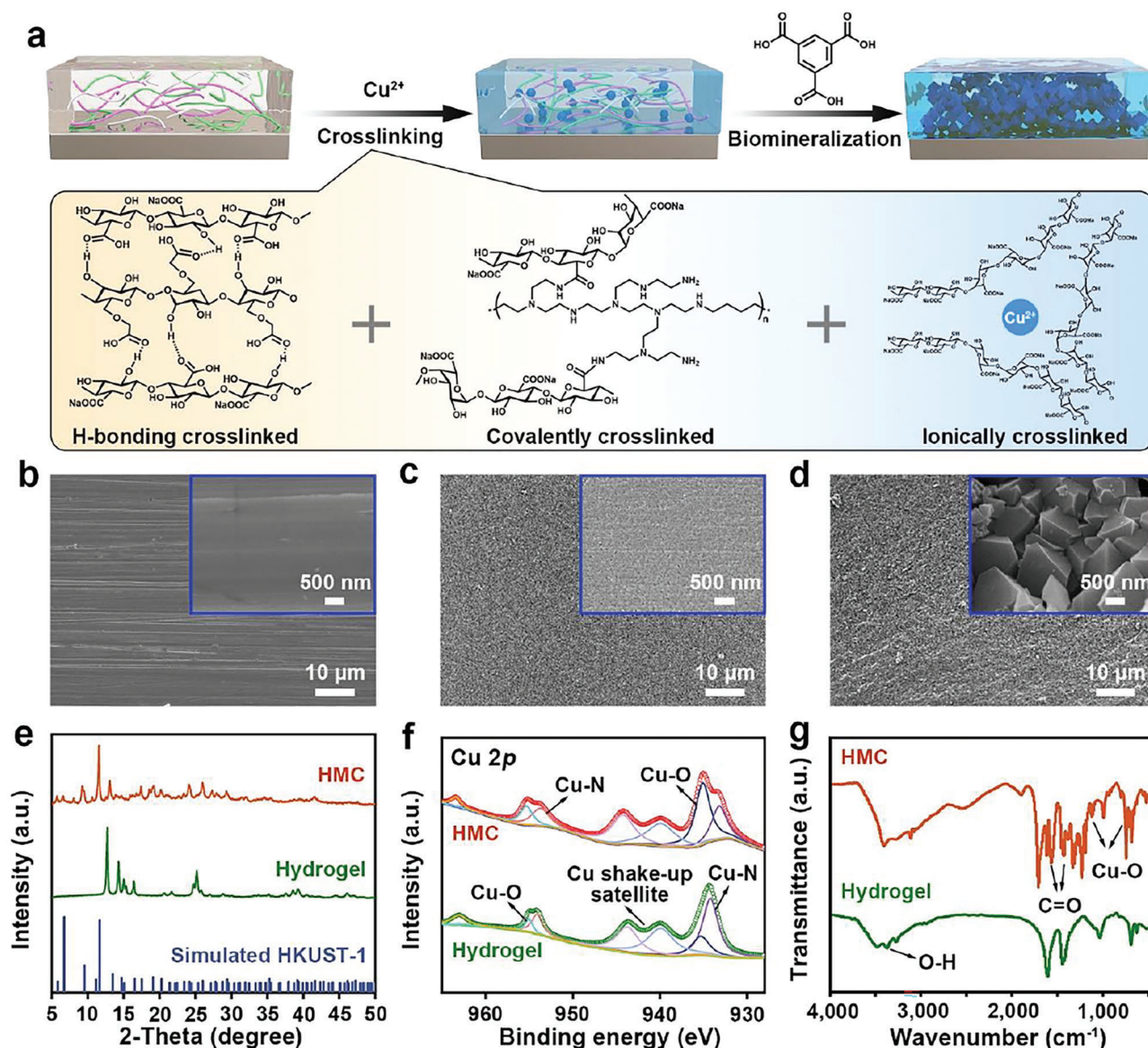


Figure 2. Fabrication process and characterizations of the HMC. a) Schematic diagram of the preparation procedures. SEM images of b) the 304 SS substrate, c) the hydrogel coating, and d) the HMC showing different preparation stages. e) XRD patterns of the hydrogel coating and HMC compared with the respective standard patterns of HKUST-1. f) High-resolution XPS spectra of Cu 2p for the hydrogel coating and HMC. g) FTIR spectra of the hydrogel coating and HMC.

Information). It is worth noting that the competitive covalent bonding of Cu^{2+} with alginate and organic ligands plays a crucial role in regulating the growth direction of MOF particles. As shown in Figure S2 (Supporting Information), the parent HKUST-1 possesses irregular nanosphere morphology. Limited by the frozen polymer chains and the chelated metal ions based on Wulff's rule,^[35] the HKUST-1 in our HMC possesses a perfect arrangement of cubic crystals with much larger particle sizes to minimize defects, fill voids between assembled MOFs, and complete densification of the processed coatings (Figure 2d). Such a unique "brick-and-mortar" configuration has a relatively higher MOF loading and intensified pore dispersion favorable for efficient antifouling and anti-corrosion activity.

The determined contents of carbon, oxygen, and copper elements are uniformly distributed throughout the interpenetrating network ($\approx 70 \mu\text{m}$), indicating that the MOF reinforcing units were interconnected with the adjacent hydrogel through bridging compositions (Figure S3, Supporting Information). Successful formation of the MOFs is confirmed by X-ray diffraction that the diffraction peaks are well in accordance with the standard patterns of the original samples (XRD, Figure 2e). The evolution of the MOFs from the well-distributed Cu^{2+} seed ions is further clarified by X-ray photoelectron spectroscopy (XPS, Figure 2f; Figure S4, Supporting Information). The high-resolution XPS spectra near the Cu 2p region show two major peaks with shake-up satellite peaks. The Cu 2p_{3/2} signal consists of two

characteristic peaks at 933.2 eV and 935.1 eV corresponding to Cu–N and Cu–O. The Cu $2p_{1/2}$ region features two peaks assigned to Cu–N (953.7 eV) and Cu–O groups (955.3 eV).^[36,37] The dramatic change of the signal intensities corroborates the competitive coordination effects among Cu²⁺, alginate, and carboxylate-based ligands in the biomineralization process. Fourier transform infrared spectroscopy (FTIR) presents that the wide characteristic peaks between 3100 $\approx$ 3700 cm^{−1} are ascribed to the overlapping of the O–H stretching vibration of SA and CMC with the N–H stretching vibration of PEI (Figure 2g). Compared with hydrogel coating, the above peaks of the HMC undergo significant red shifts, evidencing the strong intercalation of organic ligands into the system. The new peaks at 1561 and 1430 cm^{−1} are identified as the C=O asymmetric and symmetric vibrations, respectively. The corresponding frequency difference below 200 cm^{−1} reveals the bidentate bridging mode of carboxyl groups bonded with Cu²⁺ centers. In addition, the strong peaks at 1110 and 748 cm^{−1} attributed to typical Cu–O stretching vibration conclude the synthesis of HKUST-1 in the coating.^[38] Therefore, access to the HMC with the designed heterophase structural units is performed successfully according to the aforementioned characterization.

2.2. Mechanical Properties and Versatility of the HMC

The mechanical properties of the HMC were systematically investigated to evaluate the effectiveness of the heterophase “brick-and-mortar” structure at multiscale interfaces. Before incorporating the biopolymer-based additives (SA, CMC, and PEI), we first determined the optimal weight ratio of SA to CMC in terms of the tensile mechanical performance of the binary hydrogel. Compared with unary systems, the tensile tests display that the strain corresponding to the ultimate stress of the hydrogel is significantly enhanced by the addition of CMC, resulting from the cross linkage between the abundant carboxyl and hydroxyl groups of polymeric networks (Figure 3a). The tensile property of the optimal SA/CMC hydrogel (4:1) was superior to that of the other binary competitors (Figure 3b), performing a unique integration of high tensile strength (466.1 $\pm$ 5.8 kPa) and stretchability (ultimate strain up to 760%). It can be seen that the excessive CMC content could make the hydrogel less deformable and brittle due to the deterioration of intermolecular force and interfacial adhesion.^[39] According to the energy-dissipating mechanisms,^[40] we introduced PEI chains as covalently crosslinked domains to further improve the toughness of the hydrogel and the load-bearing capability of the MOF reinforcing units. Meanwhile, PEI, as a type of positively charged polyelectrolyte, can combine with negatively charged SA/CMC to form a charge-balanced surface, reducing the electrostatic interaction with pollutants to minimize the biofouling adhesion. A series of ternary hydrogels were synthesized by incremental adjustment of the PEI additive (Figure 3c). Determined via zeta potential analysis, the isoelectric point of the SA/CMC/PEI hydrogel was achieved successfully with the PEI content of 35%. As expected, the antibacterial tests display that the antibacterial adhesion of the hydrogel is consistent with the overall electrical property of the coating (insets in Figure 3c), in which the fewest bacteria are adhered to the electrically neutral surface. It should be noted that the hydration ability of the hy-

drogel is enhanced by hydrophilic hyperbranched PEI chains as identified by the low static water contact angles and strong water uptake capacity, providing a benefit for antifouling performance (Figure 3d,e). It is assumed that the polymer chains within the optimized ternary hydrogel are easier to be relaxed from the domain interior with a higher degree of chain interpenetration. More importantly, the ammonium groups of PEI chains endow the hydrogel coating with the excellent ability attached to the substrate to resist external friction (Figure 3f). Therefore, the optimal weight percentage of the mechanically superior ternary hydrogel is 59 wt.% SA, 15 wt.% CMC, and 26 wt.% PEI.

Originating from the above hydrogel scaffold, the HMC could replicate an ordered nacre-like structure to reconcile its organic hydrogel phase and inorganic MOFs phase, which presents preferable mechanical properties (Figure S5, Supporting Information). It is supposed that the embedded MOFs could bundle the polymer chains effectively to conduce the crosslinked networks tighter, realizing the energy dissipation by physical interactions, chemical bonding, and stress transfer effects.^[41] Moreover, the unique configuration can enhance the hydrophilicity to obtain excellent hydration ability (Figure S6, Supporting Information). Remarkably, the hydrogel-anchored MOF painting technique is applicable to various substrates such as organic and inorganic materials, as well as hard and soft surfaces, due to abounding interfacial adhesion interactions containing Van der Waals forces, hydrogen bonding, and hydrophobic interactions.^[42] As exemplified in Figure 3g, the HMC is successfully fabricated on a range of substrates, including metal, glass, rubber, and plastic, without special surface treatment. The extension on irregularly shaped substrates was also examined to verify the versatility of the strong adhesion to substrates. It is noteworthy that the skin of the plastic dolphin and mobula models is coated uniformly with our coatings, rendering strong evidence for the excellent mechanical properties and flexibility of the HMC (Figure 3h; Figure S7, Supporting Information). Besides the tests of changing substrate materials, other MOFs constructed from different metal or ligands can also be formulated into the HMC (Figure S8, Supporting Information), reflecting the generality and diversity of the hydrogel-anchored MOF painting technology. Through the regulation of compatible components, the HMC may become a promising choice in the efforts to overcome the incompatibility of MOFs stiffness and substrate adhesion.

2.3. Antifouling Performance of the HMC

The marine biofouling colonization on the surface involves several vital stages at micro and macro scales.^[11] Conditioning film is first formed spontaneously associated with the dynamic adsorption of biomolecules such as protein and polysaccharide, providing nutritional conditions for the subsequent biofilms from various species of bacteria. The biofilms comprising extracellular polymeric substances and immobilized bacterial cells further create a specific and heterogeneous microenvironment for the accumulation of algae and other microbial communities. The hydrophilicity, balanced charge, and easily degradable metal sources of our HMC are anticipated to effectively enhance the antifouling ability in each colonization stage. The proof-of-concept experiments were acquired using proteins, bacteria, and

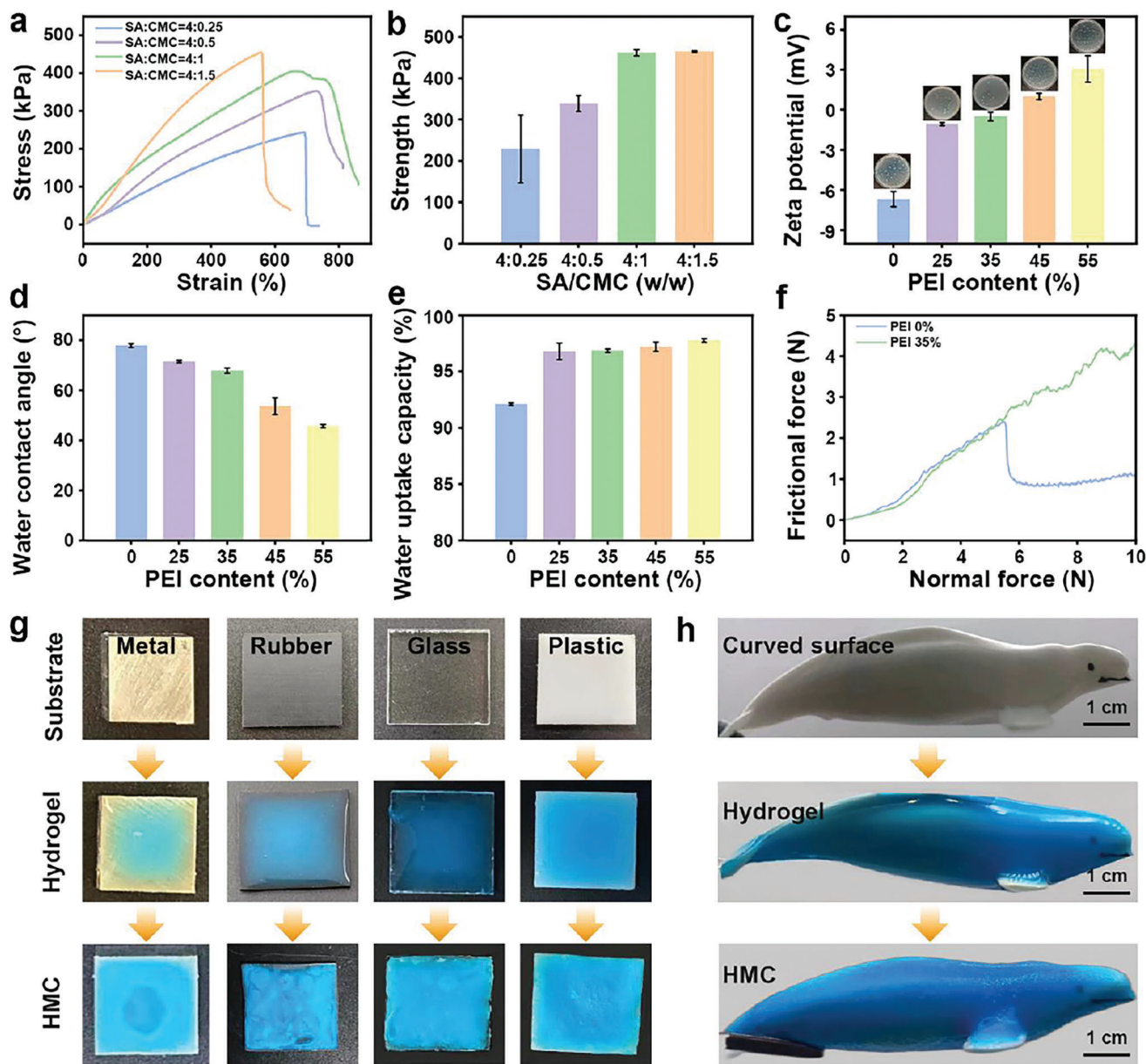


Figure 3. Assessment of mechanical performances. a) Tensile stress-strain curves of binary hydrogels with different CMC contents. b) Strength of binary hydrogels with different CMC contents. c) Zeta potentials of ternary hydrogels with different PEI contents. Insets: digital photographs showing the antibacterial behaviors on the corresponding surfaces (*Pseudomonas aeruginosa* incubation for 24 hours). d) Water contact angles of different ternary hydrogels. e) Water uptake capacities of different ternary hydrogels. f) The bonding ability of the substrate and hydrogels with different PEI contents. g) The modification of various substrates with HMC through the hydrogel-anchored MOF painting technology. h) The appearance of the coated plastic dolphins with the HMC.

algae to evaluate the antifouling performance of the as-prepared HMC.

As proteins are the critical component at the initial stage of marine biofouling, bovine serum albumin (BSA) was chosen as the model foulant to monitor the anti-adhesion property of different surfaces. Once materials are immersed in the BSA solution, the strong absorption of foulants was carried out for all the surfaces. After reaching stable absorption, PBS solution was introduced to remove the loosely bonded BSA on the surfaces.

In particular, the remnant BSAs on the hydrogel and HMC are only $0.13 \pm 0.02 \text{ mg cm}^{-2}$ and $0.13 \pm 0.01 \text{ mg cm}^{-2}$, while those on the MOF-assembled coatings without hydrogel (denoted as MC) and bare 304 SS are as high as $0.50 \pm 0.03 \text{ mg cm}^{-2}$ and $0.52 \pm 0.04 \text{ mg cm}^{-2}$, respectively (Figure 4a; Figure S9, Supporting Information). The almost complete reversible adsorption-desorption of BSA on the HMC shows the high-efficiency antifouling performance, which originates from the hydrogel-MOF heterophase structure that generates strong interfacial hydration

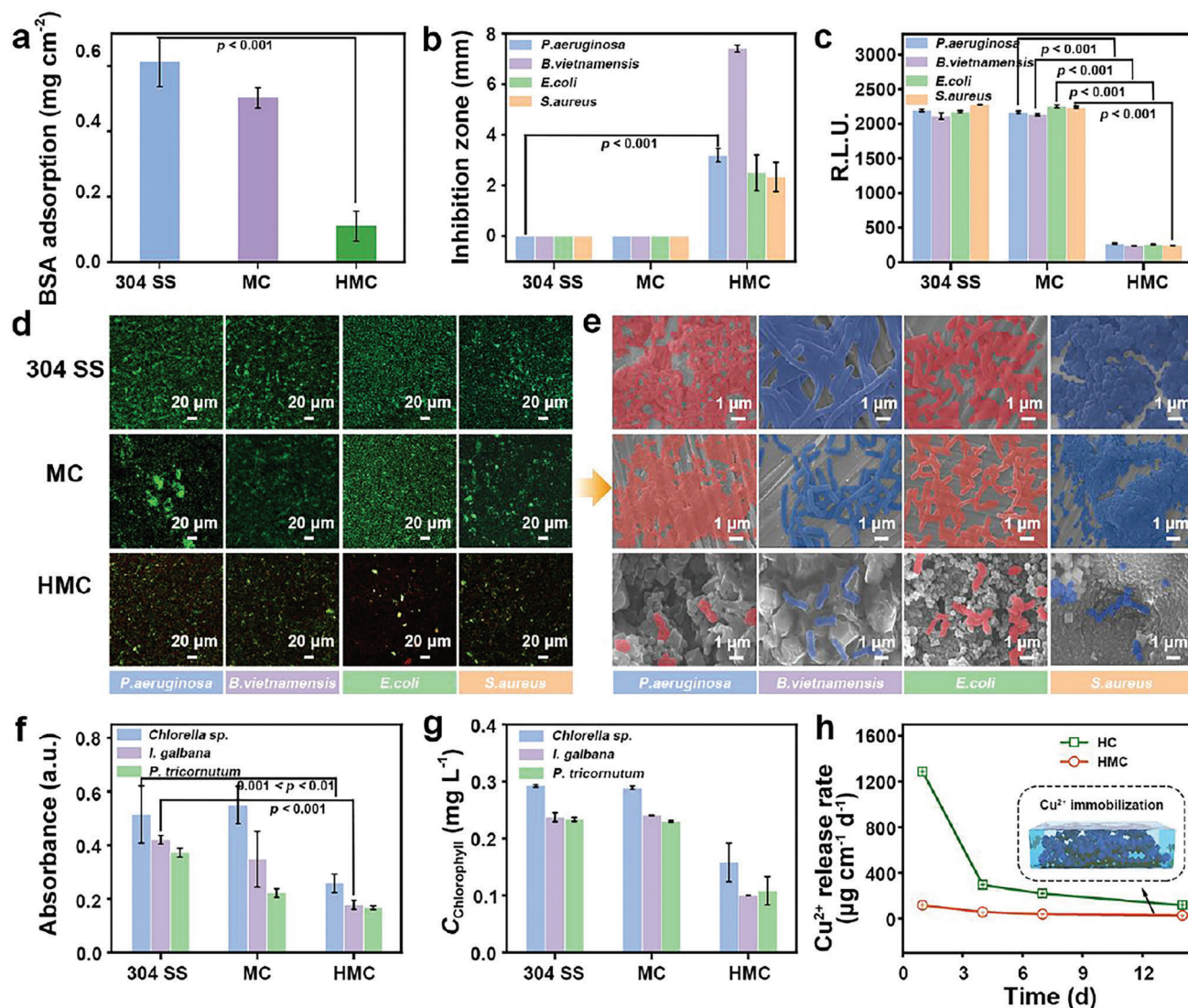


Figure 4. Antifouling properties of the coatings. a) Adsorption amount of BSA on different surfaces. Anti-bacteria properties against *P. aeruginosa*, *B. vietnamensis*, *E. coli*, and *S. aureus* of different surfaces based on b) inhibition zone diameters, c) ATP-bioluminescence, d) CLSM images of the live (green) and dead (red) bacteria, and e) SEM images. Quantitative anti-algae characterization against *Chlorella sp.*, *I. galbana*, and *P. tricornutum* based on f) algae cell biomass and g) chlorophyll concentration. h) Release rates of Cu^{2+} of the hydrogel coating and HMC after immersion in simulated seawater for 14 days. The data are presented as mean values $\pm$ SD, $n = 3$.

and steric repulsion to prevent the formation of the conditioning film.

Bacteria play an essential role in constructing biofilms at the second stage of marine biofouling. Gram-negative *Pseudomonas aeruginosa*, *Escherichia coli*, and gram-positive *Bacillus Vietnam*, *Staphylococcus aureus* were used as the model bacteria for antibacterial assays because they are representative pathogens or widely exist in marine environments. On the basis of the classical hydration property to inhibit the attachment of bacteria, the antibacterial feature of the HMC is further highlighted by its excellent Cu^{2+} -releasing ability of embedded MOF units. As examined by the agar disk diffusion method, in contrast to bare 304 SS and MC, the HMC exhibits the maximum inhibition ring diameters against *P. aeruginosa*, *B. vietnamensis*, *E. coli*, and *S. aureus* (Figure 4b). Adenosine-5'-triphosphate (ATP) can react with specific enzymes

to produce fluorescence, and the relative luminescence is proportional to the microbial biomass. In this manner, the lowest relative luminescence units (RLU) for all four types of model bacteria are observed on the HMC, which is in agreement with the above tests (Figure 4c). These results confirm that the abundant copper ions are released via HKUST-1 hydrolysis in the HMC, which can rapidly destroy the cell membrane to cause bacteria death. The antibacterial capability was directly visualized using the fluorescence staining method under a confocal laser scanning microscope (CLSM). After 7 days of incubation, the model bacteria cultured on the HMC presents notably decreased adhesion compared to that grown on the counterpart solid surfaces, indicating the formed hydration layer as a barrier improves its resistance to bacteria adhesion (Figure 4d). Moreover, the strong red fluorescence state of the HMC substantially shows the 100% lethality

of the coating to the bacteria. In comparison with the free bacteria, the adhered biofilm is much more difficult to be eliminated. SEM was utilized to evaluate the morphological changes and the integrity of biofilms (Figure 4e). Compared with numerous intact bacteria in the control groups, the HMC is attached with bare biofilms with leakage and shrunken cell membranes, indicating that the release of Cu^{2+} is crucial to broad-spectrum sterilization and antibiofilm formation capability.

To explore the antifouling diversity of the HMC, we used *Chlorella sp.*, *Isochrysis galbana*, and *Phaeodactylum tricornutum* as the typical organisms to test the growth states of algae biofilms on the coatings. The assessment of anti-algae performance was performed by immersing the samples in fresh algae suspension for 7 days under daylight/dark illumination cycles via post-rinsing treatment. Over the growth period, there is a substantial difference in algae settlement among the testing surface according to the characteristic absorbance signals, in which the HMC has significantly lower algal coverage (Figure 4f; Figure S10, Supporting Information). In addition, a similar trend of algae attachment is evidenced by the algal cell density from the amount of chlorophyll, which is remarkable indicator of the presence and effectiveness of the immobilized algal biofilm organization (Figure 4g). As expected, the CLSM observation reveals the appearance of the intense green color on both the 304 SS and MC, indicating interconnected and intact algae biofilms are firmly attached to these surfaces (Figure S11, Supporting Information). In contrast, light green fluorescence manifested in the HMC shows that our coating can avoid a large number of algae deposited on the surface. The anti-algae mechanism of the HMC originates that the generated Cu^{2+} is regarded as one of the most efficient and cost-effective anti-algae species.^[43] Cu^{2+} has negative effects on cell division, respiration, and photosynthesis by inhibiting the formation of catalase and glutathione reductase in cells.^[44,45] In addition, Cu^{2+} disrupts the morphology of both actin filaments and microtubules to depolymerize the cytoskeletal of algae.^[46] The inductively coupled plasma mass spectrometry (ICP-MS) displays that the copper ions exudation of the HMC is stable at $27.5 \mu\text{g cm}^{-2} \text{d}^{-1}$, which is sufficient to kill microorganisms ($>6.35 \mu\text{g cm}^{-2} \text{d}^{-1}$) with satisfactory environmental friendliness.^[43,47] Significantly, the heterophase “brick-and-mortar” structure of our HMC effectively solves the initial burst problem of the hydrogel system to permit a prolonged release of Cu^{2+} (Figure 4h). To summarize, the aforementioned results demonstrate that the HMC exhibited extraordinary antifouling capability to a broad spectrum of biofoulants, including protein, bacteria, and algae.

2.4. Anti-Corrosion Assessment of the HMC

Under the marine environment, metal materials highly susceptible to microbial attachment and corrosive biofilm formation inevitably undergo the MIC besides marine biofouling. It is well-acknowledged that the MIC process would lead to a local acidic microenvironment around the metal surfaces,^[48] which may trigger the rapid degradation of MOFs to enhance the antibacterial effect. Thus, our HMC with unique dense microstructures and antibacterial capability holds excellent promise for MIC inhibition. To demonstrate this potential, the pH-responsive release behav-

ior of Cu^{2+} was first assessed in the simulated seawater with different pH values (Figure 5a). The HMC presents a limited Cu^{2+} burst at pH 4 but a slow and plateaued release at pH 7. The cumulative release of Cu^{2+} in the first 2 h is $\approx 40.54\%$ and 7.85% at pH 4 and 7, respectively. The water-sensitive and acid-triggered release phenomenon indicates that the HMC can respond intelligently to the occurrence of MIC through the rapid killing of the bacteria, while remain stable in the marine environment.

The anti-corrosion performance was carried out by electrochemical measurements in simulated seawater containing corrosive *P. aeruginosa*. Electrochemical impedance spectroscopy (EIS) data of different materials were analyzed in terms of the equivalent circuit model (Figure S12, Supporting Information). In the Nyquist plots, the capacitive semicircle at a high frequency represents the charge transfer resistance of the metal/solution interface. As exhibited in Figure 5b, the HMC possesses the largest diameter of the capacitive semicircle compared with the bare substrate and hydrogel coating. In corresponding to the Bode diagram (Figure S13a, Supporting Information), the HMC features a low-frequency impedance modulus ($|Z|_{0.01 \text{ Hz}}$) of $4.29 \times 10^6 \Omega \text{ cm}^2$, which is almost 11 times as much as that of the hydrogel coating ($3.82 \times 10^5 \Omega \text{ cm}^2$) and 50 times of the 304 SS ($9.57 \times 10^4 \Omega \text{ cm}^2$). These results suggest that the designed heterophase structure of our HMC can efficiently prevent the microbial corrosion of 304 SS. This was also confirmed by a test of up to 7 days. The phase angle diagram of the HMC shows only one peak corresponding to one time constant, indicating that the corrosive medium has not diffused to the metal surface due to the remarkable barrier effect of the coating. Unfortunately, two constants in the phase angle diagram of the 304 SS manifest the occurrence of MIC (Figure S13b, Supporting Information). Correlated with the polarization curves (Figure 5c; Table S1, Supporting Information), the corrosion current density (i_{corr}) of the HMC ($0.45 \pm 0.19 \text{ nA cm}^{-2}$) is significantly lower than that of the 304 SS ($2.66 \pm 0.05 \text{ nA cm}^{-2}$) and hydrogel coating ($0.76 \pm 0.07 \text{ nA cm}^{-2}$). Our HMC demonstrates the highest corrosion potential (E_{corr} , $0.14 \pm 0.03 \text{ V}$), exceeding that of the 304 SS ($-0.62 \pm 0.19 \text{ V}$) and hydrogel coating ($-0.004 \pm 0.01 \text{ V}$). Furthermore, the pitting potentials (E_{pit}) of the 304 SS and hydrogel coating are measured at $0.20 \pm 0.07 \text{ V}$ and $0.62 \pm 0.03 \text{ V}$, respectively, inferior to that of the HMC ($0.71 \pm 0.09 \text{ V}$). The above-detailed analysis reveals that the HMC not only has the most negligible corrosion probability and slowest corrosion rate but also hinders the pitting corrosion even exposed to *P. aeruginosa*.

To determine the inhibition effect of coatings on pitting corrosion, the surface morphologies of the testing samples were observed by CLSM after removing the attached corrosion products and biofilms. It can be seen that the bare 304 SS suffers severe corrosion with abundant and deep pits, while only small and shallow pits are visible on the surfaces of the hydrogel and HMC after 7 days of immersion (Figure 5d–f). According to the statistical analysis of average and maximum pit depths, the resistance to pitting corrosion of the HMC is superior to other counterpart surfaces (Figure S14, Supporting Information). The cumulative probability implies that the ranges of pit depths for the 304 SS, hydrogel, and HMC are between 1.8 and $6.4 \mu\text{m}$, 1.3 and $2.8 \mu\text{m}$, and 0.9 and $2.0 \mu\text{m}$, respectively (Figure S15a, Supporting Information). The relationship between reduced variants and pit depths was analyzed by a double exponent (Figure S15b,

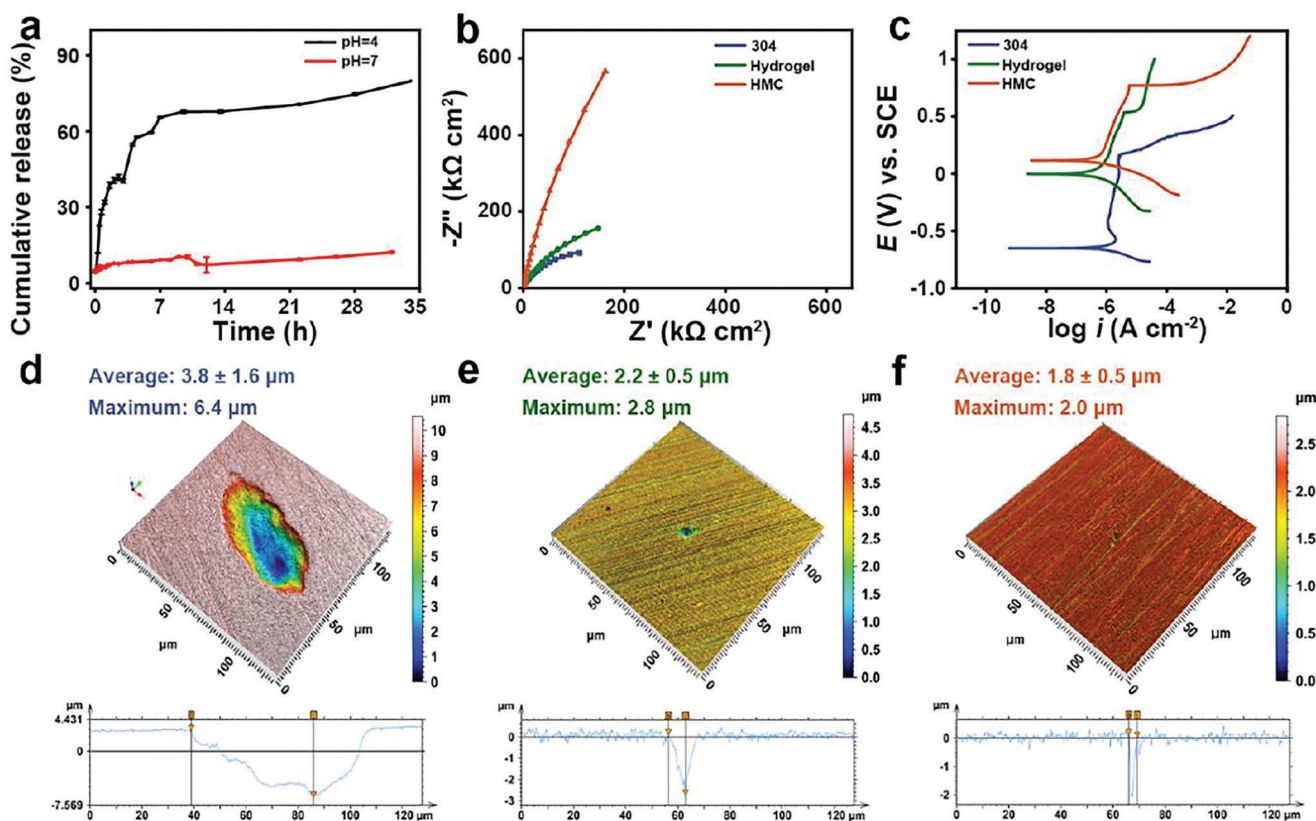


Figure 5. Corrosion behaviors of the coatings in harsh environment. a) Release profiles of Cu^{2+} from the HMC under different pH conditions. b) Nyquist plots of the 304 SS, hydrogel, and HMC after immersion in the culture media with *P. aeruginosa* for 4 days. c) Potentiodynamic polarization curves of the 304 SS, hydrogel, and HMC after immersion in the culture media with *P. aeruginosa* for 7 days. CLSM images of the d) 304 SS, e) hydrogel, and f) HMC showing the maximum pit depth after immersion in the culture media with *P. aeruginosa* for 7 days.

Supporting Information).^[49] Two linear regions in the curve of the 304 SS indicate that the pitting corrosion is obviously intensified. On the contrary, the inhibition of pitting corrosion would be achieved in the hydrogel and HMC systems due to the single linear region. Moreover, the bare metal substrate tends to form stable pits with a radius greater than 10 μm , which decreased greatly on the surface of our coatings (Figure S15c, Supporting Information). Therefore, the HMC is identified for its great potential in metal microbial corrosion inhibition, attributed to the triple synergetic effects containing the protective barrier of the dense heterophase configuration, the pH-triggered release of Cu^{2+} , and the reinforced interfacial interaction.

2.5. Extensive Drug-Loading Platform of the HMC

Besides metal ions, many biocide micromolecules are commonly used in the existing antifouling coatings to prevent secondary biofouling and corrosion recurrence. However, most biocides are prone to endogenous accumulation or biological inactivation, resulting in the decrease in antifouling and anti-corrosion performance. Due to their high specific surface area and abundant porosity, MOFs can serve as multifunctional loading platforms to separate and protect the drugs before release, achieving timely biocide release after accumulation (Figure 6a).

Brunauer-Emmett-Teller (BET) nitrogen adsorption-desorption experiments were performed to confirm the permanent porous framework of the coatings. The nitrogen sorption isotherms of the hydrogel and HMC present typical type IV adsorption curves with type H3 hysteresis loops, suggesting the mesoporous features of mesopores in the systems (Figure 6b). According to the analysis of pore width distribution (Figure S16, Supporting Information), an important structure feature is that the HMC is equipped with mesoporous apertures (25.3 nm) to connect with each other to form large cavities (93.1, 117.2, and 147.6 nm). The BET surface area of the HMC is calculated as $6.70 \text{ m}^2 \text{ g}^{-1}$ by fitting the adsorption isotherm data to the Langmuir equation, which is significantly higher than that of the hydrogel ($0.78 \text{ m}^2 \text{ g}^{-1}$). Thus, the dense construction of MOFs can ensure the rich porous structure for the drug-loading capacity of the HMC.

Here a new type of broad-spectrum biocides fabricated elaborately from natural rosin stone extracts was set as the model drug to match our HMC (Figure S17, Supporting Information). The drug-encapsulated mode of this system is delicately designed to realize the intelligent and durable release for long-term antifouling. The morphology and XRD patterns of the drug-loaded HMC are similar to the original state, indicating that the surface morphology and crystallinity of MOFs could not be affected by the encapsulated drug (Figure S18 and S19, Supporting Information).

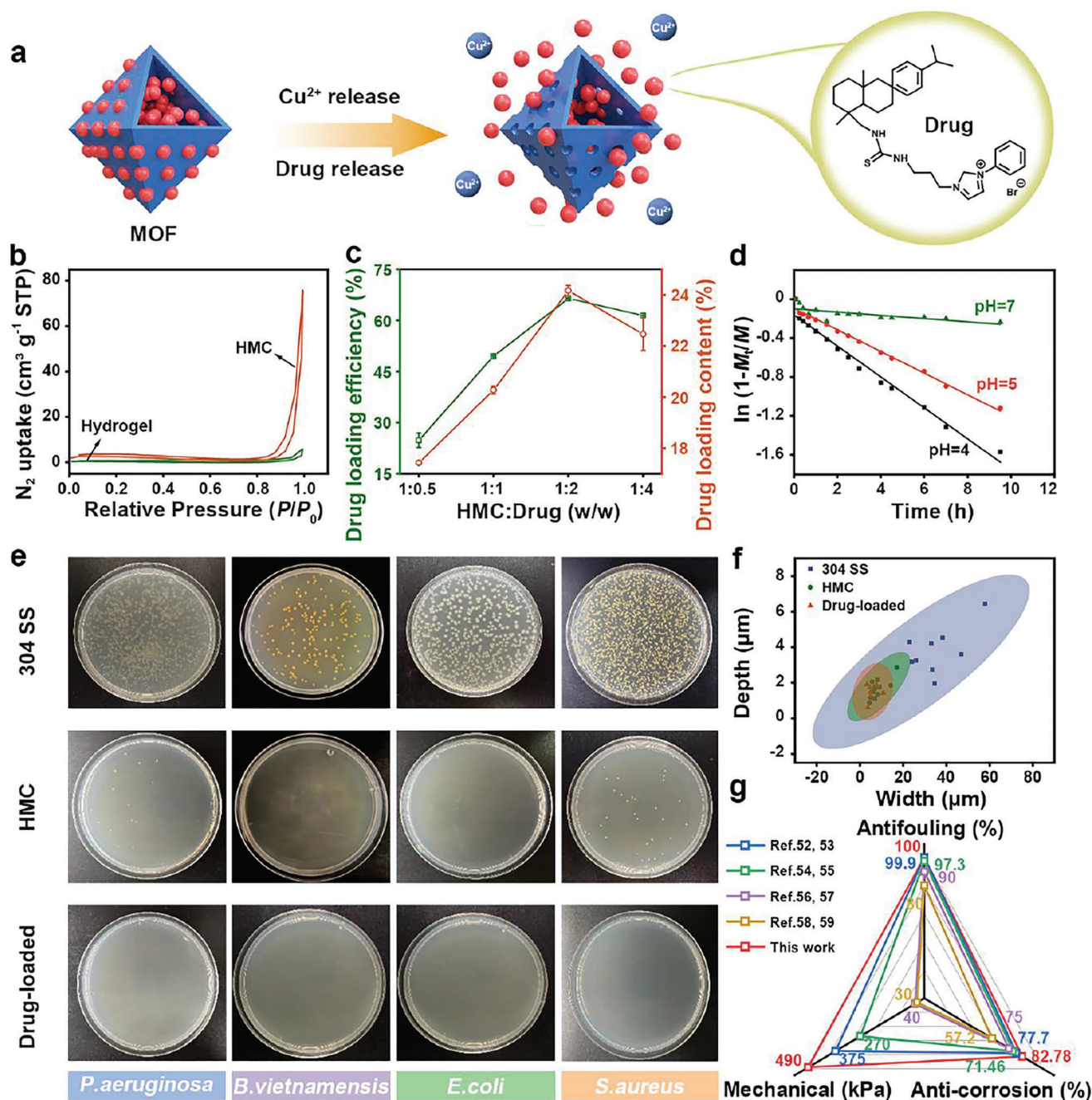


Figure 6. Management of drug-loading capability. a) Schematic diagram of the drug-loaded HMC for anti-fouling-corrosion application. b) Isotherms of nitrogen adsorption-desorption. c) Drug loading content and efficiency for different drug loading ratios in the HMC. d) Fitting of Ritger-Peppas model to drug release law. e) Coupon biofilm colonies of *P. aeruginosa*, *B. vietnamensis*, *E. coli*, and *S. aureus* treated by the 304 SS, HMC, and drug-loaded HMC in the low-nutrition environment for 7 days. f) Pits distribution after immersion in the *P. aeruginosa* solution for 7 days. g) Comparison of our HMC with some other notable reported coatings in the literature according to core requirements of marine metal protection. The antibacterial rates are used as the antifouling ability and the mechanical properties are represented by tensile strength.

These micromolecules are stably fixed in the porous structure through hydrogen bonds between the drug and the organic linkers of the MOFs. The optimal loading content and efficiency of the drug can reach $24.17 \pm 0.19\%$ and $66.48 \pm 0.04\%$, respectively, at the weight ratio of HMC to drug of 1:2 (Figure 6c; Figure S20, Supporting Information). In consideration of the local acidic mi-

croenvironment caused by fouling organisms, the release profiles of the drug were monitored under various pH conditions, conforming to the ritger-peppas model (Figure 6d) and parabolic diffusion model (Figure S21, Supporting Information).^[50] These results discern that the drug-loaded HMC would offer active and stable defense by controllable drug release aiming at the adhesion

behavior of fouling organisms (Figure S22, Supporting Information).

The antibacterial property of the drug loading manner was assessed using the dilution coating plate method. Low-nutrient and high-nutrient media were chosen to simulate the natural and harsh marine environment, respectively (Table S2–S4, Supporting Information). After the samples were incubated with bacterial medium for 7 days, there are almost no measurable levels of living bacteria survived on the drug-loaded HMC in comparison with the 304 SS and HMC whether in the natural or harsh marine environment, showing an antibacterial rate of 100% *P. aeruginosa*, *B. vietnamensis*, *E. coli*, and *S. aureus* (Figure 6e; Figure S23, Supporting Information). The data was consistent with the agar disk diffusion assay and ATP measurement, suggesting that the synergistic diffusion of both Cu^{2+} and drugs plays the role of broad-spectrum sterilization (Figure S24, Supporting Information). The resistance to proteins and algae was also strengthened after the introduction of the specific drug in the coatings (Figure S25, Supporting Information). EIS measurements were further employed to evaluate the anti-corrosion performance of the drug-loaded HMC, which shows a larger capacitive arc radius and higher low-frequency impedance modulus of $7.13 \times 10^7 \Omega \text{ cm}^2$ than those of the HMC (Figure S26, Supporting Information). The result of pits distribution also confirms that better anti-corrosion efficiency is obtained by the drug introduction in our coating (Figure 6f). Furthermore, the HMC can overcome the inherent weakness of traditional MOFs based on the sophisticated micro/nanostructure, resisting physical and chemical disturbances with longer lifetime and better durability (Figure S27–29, Supporting Information). We perform the qualitative comparison of the HMC with most reported marine coatings in accordance with the core requirements of practical metal protection (Figure 6g).^[51–59] Considering the unique advantages, we believe that our HMC potentially provides an effective portable solution for antifouling and anti-corrosion.

3. Conclusion

In summary, we demonstrate an engineered hydrogel-anchored MOF coating in accordance with a bionic “brick-and-mortar” design which overcomes the performance–processability trade-off of native MOFs and achieves robust and effective anti-fouling–corrosion application. Incorporating orderly heterophase structures and multiscale interface crosslinking interactions impart the resultant coating with enhanced strength and extrinsic toughness by virtue of sufficient energy dissipation. The surface-independent anchoring capability is contributed by multiple attractive interactions between biopolymer segments and the target substrate. The nacre-inspired coating preserves the original hydration function of hydrogel and enables a sustainable release of metal ions in response to the local acid microenvironment of biofouling and corrosion sites. We have confirmed the potential application of the HMC that possesses remarkable resistance to a wide spectrum of potential biofoulants, including proteins, bacteria, and algae, along with adequate and long-term protection against MIC. Moreover, owing to the unique porous feature, loading targeted drugs with bioactive abilities can be delicately optimized for different antifouling demands. Our study may shed light on fulfilling the technological gap for sustainable

MOF-based marine coatings, which should offer more elevating options for other advanced materials at the anti-fouling–corrosion nexus.

4. Experimental Section

Materials: Sodium alginate (SA, 90%), carboxymethyl cellulose (CMC, $M_w = 700\,000$), polyethyleneimine (PEI, $M_w = 70\,000$, 30%), copper acetate monohydrate ($\text{Cu}(\text{OAc})_2 \cdot \text{H}_2\text{O}$, 99%), and benzene-1,3,5-tricarboxylic acid (H_3BTC , 98%) were purchased from Macklin Biochemical Co., Ltd. The LIVE/DEAD BacLight Bacterial Viability Kit was purchased from Thermo Fisher Scientific Co., Ltd., Eugene, USA. Bovine Serum Albumin (BSA, 98%) was obtained from Beyotime Biotechnology Co., Ltd. *P. aeruginosa* (MCCC 1A00099) and *B. vietnamensis* (MCCC 1A00027) were obtained from Marine Culture Collection of China. *E. coli* (ATCC 25 922) and *S. aureus* (ATCC 25 923) were obtained from China Center of Industrial Culture Collection. *Chlorella sp.*, *I. galbana*, and *P. tricornutum* were obtained from Guangyu Biological Technology Co., Ltd. The 2216E medium and Luria-Bertani medium were purchased from Qingdao Hope Biotechnology Co. The F/2 medium was purchased from Guangyu Biological Technology Co., Ltd. All other reagents were of analytical grade and were used without further purification throughout the experiment. The 304 SS samples were polished with 240, 400, 600, 800, 1000, and 1200 grit abrasive papers and then cleaned in ethanol under the ultrasonic condition.

Preparation of the HMC: In a typical synthesis process, SA (0.48 g), CMC (0.12 g), and PEI (0.07 g) were first dissolved in ultrapure water (20 mL) at 50 °C for 6 h until they became homogeneous. Then, the resultant mixture was transferred and coated on the substrate by a rod applicator (thickness $\approx 70 \mu\text{m}$). Subsequently, the desired hydrogel was prepared after immersion into excessive $\text{Cu}(\text{OAc})_2 \cdot \text{H}_2\text{O}$ solution of 30 mg mL^{-1} for 6 h. Different hydrogels were obtained by adjusting the weight ratios of ternary components. Afterward, the MOFs were in situ anchored in the hydrogel by the treatment of the ethanol solution of H_3BTC with a certain concentration. Finally, the HMC was successfully acquired after being washed with ethanol three times. The drug loading in the HMC was implemented by tuning the formation process of the MOFs, in which the used biocide was added simultaneously into the ethanol solution of H_3BTC . The referenced MC without the hydrogel assistance was also prepared for comparison, which only underwent the immersion of $\text{Cu}(\text{OAc})_2 \cdot \text{H}_2\text{O}$ and H_3BTC solution.

Anti-Protein Evaluation: The standard curves of the BSA protein solutions were established by UV-vis spectra (Figure S9, Supporting Information). The testing samples were soaked in 2 mL BSA solution (0.5 mg mL^{-1}) for 24 h. Next, these samples were removed carefully with a tweezer, followed by gently washing with PBS solution to remove the weakly adhered BSA. Sterile water (1 mL) was further used to violently rinse the samples to elute BSA adhered firmly on the surfaces. The BSA absorbance was measured at $\lambda = 280 \text{ nm}$ in the collected wash solution, representing the amount of BSA adhered to the sample surface.

Anti-Bacteria Evaluation: All materials were sterilized by ultraviolet irradiation for 30 min before use. *P. aeruginosa*, *B. vietnamensis*, *E. coli*, and *S. aureus* were served as the experimental species. The single colonies of the above bacteria were isolated by plate scribing method. Note that the single colonies of *P. aeruginosa* and *B. vietnamensis* were cultured in 2216E liquid medium, while the single colonies of *E. coli* and *S. aureus* were cultured in Luria-Bertani liquid medium for 12 to 16 h at 37 °C with 180 rpm rotation to reach logarithmic growth phase. The agar disk diffusion method was used to verify the release antibacterial ability of the coatings. The testing samples were placed horizontally on the surface of the solid culture medium containing $100 \mu\text{L}$ bacterial liquid ($\approx 10^7 \text{ CFU mL}^{-1}$). The inhibition ring diameters were measured after incubation at 37 °C for 12 to 16 h. The anti-biofilm ability of the coatings was further investigated with the dilution coating plate method, ATP measurement, live/dead staining, and SEM analysis. In this assay, the low-nutrient medium (Table S2, Supporting Information) was selected to simulate the actual marine environment. The high-nutrient medium (Table S3 and S4, Supporting Information) was

chosen as the severe biofouling environment. The bacterial solution of the logarithmic growth phase was collected and diluted to 10^7 CFU mL⁻¹ with the culture medium for subsequent experiments. The testing samples were soaked in 2 mL diluted bacterial solution for 7 days, then gently washed with PBS solution to remove the weakly adhered bacteria. The dilution coating plate method was used to evaluate the antibacterial rate of the samples. Through the wiping with ATP swabs, the RLU reading of the sample surfaces was obtained by a handheld luminometer, which implied the corresponding biomass in biofilms. Cell viability in the biofilms was evaluated using the LIVE/DEAD BacLight Bacterial Viability Kit. The samples were stained with SYTO9 and PI dyes for 20 min in dark and visualized with CLSM at Ex/Em = 488/559 nm. The morphology and state of bacteria were subjected to SEM observation, in which the biofilms were fixed by 2.5% glutaraldehyde and dehydrated with gradient ethanol solutions.

Anti-Algae Evaluation: All materials were sterilized by ultraviolet irradiation for 30 min before use. *Chlorella sp.*, *I. galbana*, and *P. tricornutum* served as the experimental algae species. The algae were cultured by F/2 medium and F/2 + Si medium with visible-LED irradiation (3000 Lux) at 20 °C for 7 days (Table S5, Supporting Information). The algae solution was collected and diluted with culture medium to 10^5 cells mL⁻¹ for subsequent experiments. The standard curves of the algae were determined through UV-vis spectra. The testing samples were soaked in 2 mL diluted algae solution for 7 days. After gently washing with PBS solution, the adhesion of algae on the surfaces was directly observed through CLSM. Then, these immersed samples were rinsed with sterile water drastically to remove algae attached to the surfaces. On the one hand, the adhered level of the algae was characterized by the absorbance at $\lambda = 440$ nm according to the standard curves. On the other hand, the remaining liquid was further mixed with acetone and then sonicated for 15 min to extract chlorophyll, of which the absorbance at 660 nm was regarded as an indicator of algae attachment.

Anti-Corrosion Evaluation: Electrochemical tests were carried out in a three-electrode system containing 250 mL medium at 30 °C. The corrosive *P. aeruginosa* was selected as the experimental strain at a concentration of 10^6 cells mL⁻¹. The saturated calomel electrode, platinum sheet, and coated electrodes were chosen as reference, counter, and working electrodes, respectively. EIS and potentiodynamic polarization tests were carried out to characterize the corrosion resistance. The EIS data were recorded at a frequency ranging from 10^5 to 10^{-2} Hz with an amplitude of 5 mV. The polarization curves were recorded in a potential ranging from -0.5 to +1.2 V (versus E_{OCP}) at a scan rate of 0.333 mV s⁻¹. The pit depth was measured by CLSM after the removal of corrosion products, biofilms and coatings. To better understand the generation and development of pitting corrosion, the theory of pitting growth probability was introduced into the analysis.

The formula of cumulative probability was used to analyze the random distribution probability of pitting.

$$F = \exp \left\{ - \exp \left[- \frac{(d_i - \mu)}{\alpha} \right] \right\} \quad (1)$$

$$F = \frac{n}{N + 1} \quad (2)$$

where F represents the cumulative probability, d_i , μ , and α are the pit depth, central parameter, and scale parameter, respectively. n and N refer to the sample sequence number and the number of all pits.

The formula of Gumbel distribution was used to analyze the distribution of pit depth.

$$-\ln(-\ln F) = \frac{d_i - \mu}{\alpha} \quad (3)$$

The formula of double exponent was used to analyze the probability of the largest pit depth (Pit_{max}).

$$Pit_{max} = \mu + \alpha \ln S \quad (4)$$

$$P = 1 - \exp \left\{ - \exp \left[\frac{-(R_{pit} - [\mu + \alpha \ln S])}{\alpha} \right] \right\} \quad (5)$$

where P is the pit size probability, R_{pit} is the radius of pits, and S is the area of the coupons ($S = 1$ cm²). Therefore, the formula can be simplified as follows.

$$P = 1 - \exp \left\{ - \exp \left[- \frac{(d_i - \mu)}{\alpha} \right] \right\} \quad (6)$$

Drug-Loading Platforms Evaluation: Here, the natural rosin stone extract was selected as the model drug encapsulated in the HMC. The drug loading content (DLC) and drug loading efficiency (DLE) of the coatings were explored according to the following expressions:

$$DLC = \frac{W_d}{W_c} \times 100\% \quad (7)$$

$$DLE = \frac{W_d}{W_f} \times 100\% \quad (8)$$

where W_c is the total weight of the coating, W_d represents the loading weight of the drug in the HMC, and W_f refers to the weight of the drug in the feed.

The drug release pattern was analyzed Ritger-Peppas model and parabolic diffusion model.

$$M_t/M_\infty = kt^n \quad (9)$$

$$(M_t/M_\infty)/t = kt^{-1/2} + b \quad (10)$$

Where M_t/M_∞ is the release ratio of the drug, t is the release time, k is the kinetic constant, n is the diffusion exponent, and b represents a constant.

Characterization: The surface morphology of the coatings was characterized by a field emission scanning electron microscope (FE-SEM, MIRA LMS TESCAN, Czech Republic). The crystal structures of coatings were determined by X-ray diffraction (XRD) on a Rigaku SmartLab SE diffractometer using Cu-K α radiation (40 kV, 40 mA) in the 2θ range 5–50°. The chemical composition of the samples was determined with the means of X-ray photoelectron spectroscopy (XPS, K-Alpha, Thermo Fisher Scientific, USA) and Fourier transform infrared spectroscopy (FTIR, Nicolet iS20, Thermo Fisher Scientific, USA). The micro-material tensile testing system (MMT-101N, Shimadzu, Japan) was used to evaluate the mechanical behaviors of the coatings. The adhesion between coating and substrate was obtained via Micro Scratch Tester (NHT2+MST, CSM, Switzerland). The zeta potential was tested by the Malvern Zetasizer Nano ZS Instrument (ZEN3600, Malvern Instrument, UK). The contact angles were probed with a contact angle measurement machine (OCA 25 machine, Data-Physics, Germany). Moreover, the final contact angle of each sample was calculated by measuring three contact angles of different positions and taking the average. The concentration of target substances was measured using UV-vis spectroscopy (Genesys 50, Thermo Fisher Scientific, USA). Confocal images were collected by a confocal laser scanning microscopy (CLSM, Model C2 Plus, Nikon, Japan). The electrochemical tests of coated electrodes were recorded using an electrochemical workstation (Reference 600+, Gamry instrument, USA). The release of Cu²⁺ from the coatings was assessed by an inductively coupled plasma optical emission spectrometer (ICP-OES, ICAP PRO, Thermo Fisher Scientific, USA). The pore textural natures of the samples were analyzed through nitrogen adsorption-desorption isotherms using a surface area analyzer (ASAP2460, Micromeritics, USA). The thermal decomposition behaviors were evaluated by a thermogravimetric analyzer (DSC/DTA-TG, STA 449 F5 Jupiter, NETZSCH, Germany).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

Z.Y. and X.L. contributed equally to this work. This work was financially supported by the National Key Research and Development Program of China (Grant No. 2022YFB3808800, D. X.), the National Natural Science Foundation of China (Grant No. U2006219, D. X.), National Postdoctoral Program for Innovative Talents (Grant No. BX20220059, X. L.), China Postdoctoral Science Foundation (Grant No. 2022M720677, X. L.), Liaoning Provincial Natural Science Foundation of China (Grant No. 2023-BS-052, X. L.), and the Fundamental Research Funds for the Central Universities (Grant No. N2302015, X. L.).

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

antifouling coatings, corrosion inhibition, mechanical roughness, metal-organic frameworks, nacre-inspired structures

Received: May 29, 2023

Revised: June 27, 2023

Published online: July 26, 2023

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